

Machine Learning

Task 1

Learning Goals:

- Apply basic Python programming skills using loops and lists.
- Understand the levels of intelligence in data science and their real-world applications.
- Perform basic data preprocessing including handling missing values and encoding categorical data.

Task 1: Given a list `arr = [2, 0, 1, 2, 0, 1]`, write a Python script using a while loop to sort the numbers so all **0s** come first, then **1s**, and finally **2s**

Task 2: In Machine Learning, we sometimes need to generate thousands of random data points to test our models. Write a script that uses a for loop to generate and print 10 random numbers between 1 and 100.

Task 3: Choose a unique industry and explain how it uses the four levels of intelligence: Statistics, AI, Machine Learning, and Deep Learning.

Task 4: Load the given dataset and perform basic data preprocessing steps including handling missing (NA) values, imputing numeric and categorical data, and encoding categorical variables using appropriate techniques. The goal is to prepare a clean, machine-learning-ready dataset and understand the importance of preprocessing in ML pipelines.